Keru Wang

Human–AI Interaction · Proactive Multimodal Agents · Mixed Reality · HCI

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Summary

Fifth-year Computer Science Ph.D. candidate at NYU, advised by Prof. Ken Perlin, studying how interactive systems can help people think, create, and work together. With AI, I build proactive multimodal agents that understand people in context: inferring an individual's latent, evolving intent, and supporting groups through creativity support and by connecting fragmented knowledge across members. In mixed reality, I build collaborative XR environments, generative 3D authoring tools, and robotic and shape-changing interfaces for physical interaction.

Publications and demos at UIST, DIS, VRST (Best Paper), SIGGRAPH, TEI, and ICRA; recent work under review at CHI and ICLR.

AI/Agents: Agent design, multi-agent systems, tool and skill authoring, prompt engineering, multimodal (vision, speech), RAG, LLM APIs.
Engineering: Python, JavaScript, React, C/C++, Azure, WebGL, WebXR, Unity, Unreal Engine, Arduino.
Methods: Mixed-methods research, controlled and field studies, thematic analysis, surveys, statistics, agent benchmark and evaluation design.

Professional Experience

Microsoft Research, Deep Learning Group
A. Banburski-Fahey, N. Jojic

Research Intern — *May 2026 – Aug. 2026*

- Led research on a shared-memory agent system that connects fragmented knowledge across a community and surfaces it when it matters.
- Built and deployed a multi-agent LLM system with multimodal retrieval in a two-week, 46-user field study; first-authored a CHI 2027 submission.
- Contributed to research on long-horizon code agents that improve through validated skill optimization (ICLR 2027 submission).

Part-time Research Intern — *Oct. 2026 – present*

- Extending the shared-memory agent system toward proactive recall that surfaces and acts on relevant memories before users ask.

New York University, Future Reality Lab
Supervisor: Prof. Ken Perlin

Student Researcher — *Sep. 2022 – present*

- Leading research on proactive multimodal agents, from creativity support in group ideation to inferring users' latent, evolving intent.
- Developed a WebXR-based collaborative mixed reality platform with a customized shader and multimodal interfaces, deployed in research projects and graduate-level VR courses at NYU and KAIST. | Github | Demo
- Built 3D content creation tools in VR using data-driven generative model, with the project receiving the Best Paper Award at ACM VRST 2024.
- Developed robotic interfaces for mixed reality experiences, enabling haptic feedback and the actuation of passive objects. | Demo1 | Demo2

MIT Media Lab, Tangible Media Group
Supervisor: Prof. Hiroshi Ishii

Research Intern — *Jun. 2021 – Dec. 2021*

- Led the design and development of a new version of SandScape for permanent exhibition in the MIT Museum—a tangible interface for designing and exploring landscapes by manipulating sand with real-time simulations projected onto the sandbox. | Gallery
- Reconstructed the sandbox's 3D mesh in real time with openFrameworks (C++) and libigl, and simulated elevation, slope, and drainage. | Github

MiSynth
CEO: Senaida Ng

Software Engineer — *Sep. 2020 – Mar. 2021*

- Built an EEG-to-music streaming pipeline and web interfaces (JavaScript) for a brain-computer interface music startup. | Website
- Won the Most Impact Award in NYC Media Lab and ASCAP Immersive Studio Challenge. | News

Harvard Medical School
Supervisor: Dr. Michael Chou

Data Visualization Designer — *Sep. 2020 – Jan. 2021*

- Modeled and visualized the impact of mask-wearing and social distancing on the spread of COVID-19 in a web-based JavaScript tool.

Selected Publications

Under Review

[U.1] **Keru Wang**, Michael Xu, Ken Perlin, Nebojsa Jojic, Andrzej Banburski-Fahey. *SEAM: Situated Executable Associative Memory for a Co-located Community.* Under review at CHI 2027.

[U.2] **Keru Wang**, Pincun Liu, Saining Xie, Ken Perlin. *An Investigation of Proactive AI Agents for Reducing Idea Stagnation in Group Brainstorming.* Under review.

[U.3] **Keru Wang**, Pincun Liu, Saining Xie, Ken Perlin. *Should I Speak Up? Aligning Proactive AI Interventions with Human Expectations in Group Brainstorming.* Under review.

[U.4] Yifan Wang, Hao Cheng, Xiaomin Li, Yuexing Hao, Hemanth Neelgund Ramesh, Dongwon Jung, Hao Tang, **Keru Wang**, Chenliang Zhou, Qianhui Wu, Wenlin Yao, Ananth Grama, Andrzej Banburski-Fahey, Baolin Peng, Jianfeng Gao, Jaron Lanier. *Certified Long-Horizon Code Agent Evolution via Validation-Gated Skill Optimization.* Under review at ICLR 2027.

Published

[C.1] Hushen Hu, **Keru Wang**, Yuli Shao, Jan Plass, Zhu Wang, Ken Perlin. *Generative Terrain Authoring with Mid-air Hand Sketching in Virtual Reality.* ACM VRST, 2024. 🏆 ***Best Paper Award***

[C.2] **Keru Wang**, Zhu Wang, Ken Nakagaki, Ken Perlin. *"Push-That-There": Tabletop Multi-robot Object Manipulation via Multimodal 'Object-level Instruction'.* ACM DIS, 2024.

[C.3] Zhenyi He, **Keru Wang**, Yushan (Brandon) Feng, Ruofei Du, Ken Perlin. *GazeChat: Enhancing Virtual Conferences with Gaze-aware 3D Photos.* ACM UIST, 2021.

[C.4] **Keru Wang**, Zhu Wang, Karl Rosenberg, Zhenyi He, Dong Woo Yoo, Un Joo Christopher, Ken Perlin. *Mixed Reality Collaboration for Complementary Working Styles.* ACM SIGGRAPH Immersive Pavilion, 2022.

[C.5] Yuhan Wang, **Keru Wang**, Zhu Wang, Ken Perlin. *Robotecture: A Modular Shape-changing Interface Using Actuated Support Beams.* ACM TEI, 2025.

[C.6] Yi Wu, Agnieszka Roginska, **Keru Wang**, Zhu Wang, Ken Perlin. *A Spatial Audio System for Co-located Multi-participant Extended Reality Experiences.* ICAD, 2024.

[C.7] **Keru Wang**, Yi Wu, Pincun Liu, Zhu Wang, Agnieszka Roginska, Qi Sun, Ken Perlin. *Audio-influenced Pseudo-patics: A Review of Effects, Applications, and Research Directions.* AM.ICAD, 2025.

[C.8] **Keru Wang**, Pincun Liu, Ken Perlin. *Exploring Tangible and Graphical Interfaces for Large Virtual Object Manipulation*. ACM SIGGRAPH I3D, 2025.

[C.9] Lou De Bel-Air, Luca Morando, Ruitao Chen, **Keru Wang**, Benjamin Jarvis, Charbel Toumieh, Yang Zhou, Ken Perlin, Dario Floreano, Giuseppe Loianno. *Flying Together: Human-Guided Immersive Shared Control for Aerial Robot Teams in Unknown Environments*. IEEE ICRA, 2026.

Honors & Awards

2024 **Best Paper Award**, ACM VRST
2021 **Dean's Undergraduate Research Fund**, NYU Courant Institute
2021 **Most Impact Award**, NYC Media Lab and ASCAP Immersive Studio Challenge
2019 **HKSAR Government Scholarship**, City University of Hong Kong

Education

New York University, Courant Institute

Ph.D. Candidate in Computer Science, Supervised by **Prof. Ken Perlin** — *2022 – Present*

New York University

B.A. in Computer Science (Courant) and Interactive Media Arts (Tisch), Dean's List — *2019 – 2022*

Academic Service

Reviewer (2022 – present): IEEE TVCG, VR, ISMAR; ACM CHI, UIST, SIGGRAPH Asia, IUI, ETRA, TEI, ICMI; ECIS.
Workshop Organizer: XRMemory: The Future of Memory Capture and Replay Through XR and AI, IEEE VR 2025.

Talks and Exhibitions

2026 **A Shared Memory for a Community That Connects, Reveals, and Acts**, Microsoft Research
2025 **Talk on VR Meditation with Generative Sketching and Audio**, ACM SIGGRAPH Asia — *Hong Kong*
2024 **Talk on Mixed Reality Workspace**, ACM SIGGRAPH Asia — *Tokyo, Japan*
2024 **Guest Lecture on Interdisciplinary Practice in Art, Design, and Research**, Simon Fraser University — *Vancouver, Canada*
2022 **Permanent Exhibition for SandScape**, MIT Museum — *Boston, USA*
2022 **Panel discussion on VR for Education**, ACM SIGGRAPH — *Vancouver, Canada*

Teaching Experience

NYU Courant Institute

Teaching Assistant — *2020 – present*

- Graduate: Computer Graphics (Fall 2024–2026), AI Graphics (Spring 2024), Virtual Reality (Spring 2022).
- Undergraduate: Computer Graphics (Fall 2021–2023), Basic Algorithms (Fall 2020, Spring 2021, Spring 2023).